



Summary

- An improvement of the new LVPS design is the segmentation of the power supply into 8 independent bricks, which enhances the reliability of the readout systems, and minimizes the impact of a possible failure during data taking
- The design of the brick was improved both by increasing efficiency and reducing the signals for controlling the brick turning on/off
- All the components were procured from single (limited) batch(es) to perform the tests according to the guidelines from RETF2020 and the tests were performed on all of them
- The originally selected SI8920 isolation amplifier showed some fluctuations during SEE/TID tests and was initially replaced by HCPL7800 for pre-production
- Further tests showed stable performance of SI8920 in a specific voltage range and the final design of the brick was slightly changed accordingly to have stability with SI8920